# W04 报告

## 1. 测试范围与环境

本文核对本地向量库、三路召回、端到端探针，以及五个 Dart 包的默认单测覆盖率。默认套件排除 `integration`，不加载模型和数据集。冻结的 25 条证据集没有改。

环境见 `logs/env_w4.txt`：

- Dart 3.13.4（stable）
- Java 17.0.4
- Python 3.13.2（`engine/.venv`）
- `C:\Windows\System32\winsqlite3.dll` 存在。容量探针读到 SQLite 3.53.4，WAL
- chromadb 1.5.9，见 `logs/chroma_baseline.txt`
- `datasets/samples/sample.pdf`、`sample.docx`、`sample.png` 和 COCO 披萨图都在

判定以日志里的退出码、计数和耗时为准。终端截屏在第 5 节，路径相对本报告。若干 `e2e_*.txt` 的 `query=` 行是 PowerShell 把 Dart 的 UTF-8 输出按系统代码页解开的，命令注释行仍是原来的中文问句。

机器网卡没有关掉。离线核对是三个 `lib/` 的 grep，加上一次只拦 Dart 和 Python 出站的防火墙规则。规则已经删除。见 OFF-01。

## 2. 用例表

| 用例 ID | 命令 | 期望 | 实际 | 判定 | 证据 |
|---|---|---|---|---|---|
| ANA-01 | 三个新包各自 `dart analyze` | 0 issue | 三个包都是 `No issues found!`，退出码 0 | 通过 | `logs/dart_analyze.txt`；`screenshots/ANA-01-dart-analyze.png` |
| UNIT-01 | 三包 `dart test --exclude-tags integration` | 默认套件全绿 | 2026-10-26：存储 `00:02 +23`，检索 `00:02 +9`，编排 `00:03 +23`，见 `logs/coverage.txt`。2026-10-27 只重跑存储和检索：`00:03 +24`、`00:02 +11`。编排没有重跑 | 通过 | `logs/coverage.txt`；`logs/coverage_20261027_storage_retrieval.txt` |

| 用例 ID | 命令 | 期望 | 实际 | 判定 | 证据 |
| --- | --- | --- | --- | --- | --- |
| COV-01 | 五包 `--coverage` 后 `--report-on=lib` | 新三包 ≥ 85%；解析不低于 80%；嵌入不低于 85% | 存储 574/666 = 86.19%，检索 158/164 = 96.34%，编排 272/298 = 91.28%，解析 979/1091 = 89.73%，嵌入 1250/1458 = 85.73% | 通过 | 覆盖率报告.md；`coverage/*.lcov.info`；`screenshots/COV-01-coverage.png` |
| H1 | Chroma 与 Dart Top-5 对照 | 125 个位置一致率 ≥ 96% | `agreed=125, agreement=1.000000, passed=1` | 通过 | `logs/storage_equivalence.txt`；`screenshots/H1-storage-equivalence.png` |
| CAP-01 | `packages/core_storage/bin/capacity_probe.dart` 写 1 万条 | ≤ 600 ms | `write_ms=444`。SQLite 3.53.4，WAL | 通过 | `logs/storage_capacity.txt`；`screenshots/CAP-02-capacity.png` |
| CAP-02 | 同上，384 维查询中位 | ≤ 60 ms | 中位 44 ms，最小 41，最大 50。`EXPLAIN` 走 `idx_chunk_space` | 通过 | `logs/storage_capacity.txt`；`screenshots/CAP-02-capacity.png` |
| CHR-01 | `engine/chroma_baseline.py` | 记下 chromadb 版本、Hit@5、MRR@10。这是导出向量上的对照，不是产品检索器 | chromadb 1.5.9，`hnsw:space=cosine`。Hit@5 0.880000 [0.700442, 0.958332] (22/25)，MRR@10 0.795714。对 numpy 125/125 | 通过 | `logs/chroma_baseline.txt`；`screenshots/CHR-01-chroma-baseline.png` |
| E2E-01 | `index_probe` 建 `reports/week4/.index/evidence.sqlite` | 样本和 COCO 小样入库 | 样本 3/3，97 块，12045 ms。COCO 50/50，50 块，7048 ms。合计 147 块、53 个文档，`index_ms_total=19093`。`keyword_terms=0` | 通过 | `logs/e2e_build.txt`；`logs/e2e_samples.txt`；`screenshots/E2E-01-build.png` |
| E2E-02 | 同一库再跑一次 `index_probe` | 内容未改则全部复用 | 样本 `reused=3`，COCO `reused=50, written=0`。统计仍是 147/53。`index_ms_total=6873` | 通过 | `logs/e2e_rebuild.txt`；`screenshots/E2E-02-rebuild.png` |

| 用例 ID | 命令 | 期望 | 实际 | 判定 | 证据 |
| --- | --- | --- | --- | --- | --- |
| E2E-03 | 临时目录三份 Markdown，改 `a.md` 后再索引 | 未改的文件复用；改过的文件替换旧行，文档数不增加 | 第一次 3 个文档、378 块、16130 ms。追加一句后 `written=1, reused=2,` 85 块，5117 ms。`stats_documents=3, stats_chunks=379` | 通过 | `logs/e2e_incremental.txt`；`screenshots/E2E-03-incremental.png` |
| E2E-04 / H3 | `search_probe --top 10 --query "工具靠什么判断文件类型？"` | 问句不含文件名。Top-5 出现目标文件，并给出页和偏移。原文件在磁盘上 | Top-10 都是文本。`sample.pdf` 在第 1、2、4、5 名，`sample.docx` 在第 3 名。第 4 名 `page=1, start=617, end=710`，片段含 `missing or misleading extension`（证据跨度 679–710 落在这块内）。两个目标文件都存在。这次记录是权重 0.2。2026-10-27 用 0.49 重测同一句：`sample.pdf` 仍在第 1、2 名，第 3 名是 `sample.png`（`clipShared`），第 5 名是 `000000463730.jpg, 617–710` 落到**第 6 名**，Top-5 里不再有这句。H3 的判据是「Top-5 出现目标文件并给出页和偏移」，第 1 名的 `423–517` 仍满足，判定不变 | 通过 | `logs/e2e_query.txt`；`logs/e2e04_rank_after_reweight_20261027.txt`；`logs/env_w4.txt`；`screenshots/E2E-04-h3-query.png` |
| E2E-05 | 同一进程第二条问句「这份说明在能够被完全信任之前，还要求人做什么？」 | 再出一份带出处的 Top-K | 第 1 名 `sample.pdf, page=1, start=160, end=253`，片段含 `requires human review。` `encode_ms=67` | 通过 | `logs/e2e_query.txt`；`screenshots/E2E-05-second-query.png` |

| 用例 ID | 命令 | 期望 | 实际 | 判定 | 证据 |
| --- | --- | --- | --- | --- | --- |
| E2E-06 | `--extension .pdf`、`--path-prefix sample.pdf`、`--modified-after` | 扩展名和路径只留 pdf；过去的时间仍有命中；2099 年之后为 0 | `.pdf` 与 `sample.pdf` 的前几名都是 `sample.pdf`。`2020-01-01` 的第 3 名仍是 `sample.docx`。`2099-01-01` 为 `hits=0` | 通过 | `logs/e2e_filters.txt`；`screenshots/E2E-06-filters.png` |
| E2E-07 | 问「哪张图里有人在厨房做披萨？」，以及 `--extension .jpg` | Top-5 出现 `000000397133.jpg` | 不限类型时 Top-10 全是 `sample.pdf` / `sample.docx`。只留 `.jpg` 时五条都是图，第 1 名是 `000000403385.jpg`，披萨图不在这五条里。英文证据句只走 clipShared 时该图排第 1。2026-10-27 用 0.49 重测本行判据：不限类型时 `000000397133.jpg` 在第 3 名，加 `--extension .jpg` 时第 1 名就是它，Top-5 出现目标文件这一条已满足。本行仍按 2026-10-25 记录判「未通过」；重测见 `logs/e2e07_rerun_20261027.txt`，不改这一行 | 未通过 | `logs/e2e_query.txt`；`logs/e2e_filters.txt`；`logs/e2e07_clip_diagnosis.txt`；`logs/e2e07_rerun_20261027.txt`；`screenshots/E2E-07-image.png` |
| E2E-08 | 同一问句比较 `--arm A5` 与 `--arm A4` | 邻域不改名次和偏移；A5 的片段带上相邻块 | 两边第 4 名都是 `score=0.007813, page=1, start=617, end=710`。A5 片段多出前一块的 `interface, making Tika useful`。这次记录是权重 0.2，命令是 `--top 5`。2026-10-27 用 0.49 重测同一句的 A4 与 A5：两边第 4 名都是 `sample.docx`（306–426），第 5 名都是 `000000463730.jpg`，617–710 都在第 6 名，仍不进 `--top 5` 的输出。A5 与 A4 | 通过 | `logs/e2e_neighborhood.txt`；`logs/e2e04_rank_after_reweight_20261027.txt`；`screenshots/E2E-08-neighborhood.png` |

| 用例 ID | 命令 | 期望 | 实际 | 判定 | 证据 |
| --- | --- | --- | --- | --- | --- |
| | | | 在这句上名次和偏移仍相同，邻域不改名次这条成立；但被比较的第 4 名已不是 617–710，这条用例的比对对象随权重一起变了 | | |
| E2E-09 | index_probe --cancel-after 1，目录里 a.txt、b.txt、c.txt | 已写入的文件留下，后面的文件不写 | written=1, cancelled=true, stats_documents=1。检索命中 a.txt。问第二句仍返回这一行，因为库里只有一行，且没有分数下限 | 通过 | logs/e2e_cancel.txt; screenshots/E2E-09-cancel.png |
| E2E-10 | 目录里一份好 txt 和一份坏 docx | 坏文件记失败，好文件仍入库，整批退出码 0 | ingest_failed=1, written=1，报告 failed=0。检索命中 good.txt | 通过 | logs/e2e_cancel.txt; screenshots/E2E-10-bad-file.png |
| E2E-11 | 空库上 search_probe | hits=0 | 清库前的空库查询 hits=0 | 通过 | logs/e2e_clear.txt; screenshots/E2E-11-empty.png |
| E2E-12 | 写入后再 --clear，然后查询 | 块数从 1 到 0，查询为空 | clear_before_chunks=1, clear_after_chunks=0，随后 hits=0 | 通过 | logs/e2e_clear.txt; screenshots/E2E-12-clear.png |
| DOC-01 | index_probe --path docs 两次，再搜「产品侧用系统 SQLite 保存向量」 | 第一次写入，第二次复用。中文问句能命中 ADR | 9 个文件、2870 块，第一次 110092 ms，第二次 reused=9，5481 ms。keyword_terms=14915。第 1 名是 ADR 0004，start=1304，end=1336，通道 documentText,keyword | 通过 | logs/e2e_docs.txt |
| H2 | eval_retrieval.py 六臂 dev，然后 A4 holdout 一次 | 融合的 Hit@5 与 MRR@10 不低于任一单通道。没有增益就照实写，不改权重 | dev 上 A4 与 A1 同为 10/15、MRR 0.622222，高于 A0 和 A2。没有正增益。holdout A4 为 5/10。parameters_changed_after_freeze=0，holdout 只打了 1 次 | 通过（无正增益） | 检索精度基准报告.md; logs/retrieval_bench_dev.txt; logs/retrieval_bench_holdout.txt; screenshots/H2-dev.png; |

| 用例 ID | 命令 | 期望 | 实际 | 判定 | 证据 |
| --- | --- | --- | --- | --- | --- |
| | | | | | screenshots/H2-holdout.png |
| OFF-01 | 三个新包 lib/ 搜索 HttpClient\|Socket\|import .http\|package:http，并在出站被拦时对已建好的库做查询 | 无匹配；外部地址失败时，本地查询仍有命中；规则事后不留 | 扫描 40 个 dart 文件，matches=0。unshare -rn 里网络与 DNS 不可达，索引与语料可读。防火墙：加规则前 223.5.5.5:443 能连；规则生效时 Python 为 WinError 10013，Dart 出站失败。同时 search_probe --arm A5 问「工具靠什么判断文件类型？」search_exit=0，第 1 名 sample.pdf，第 4 名 start=617 end=710。删规则后再 show 为 No rules match，该地址恢复。第 4 名是权重 0.2 的记录。2026-10-27 用 0.49 重测同一句：617–710 在第 6 名，第 1 名仍是 sample.pdf（423–517）。断网判据取的是「本地查询仍有命中」，不依赖第 4 名。防火墙那一段没有重跑，权重与规则无关 | 通过 | logs/offline_check.txt；logs/offline_namespace.txt；logs/offline_firewall.txt；logs/e2e04_rank_after_reweight_20261027.txt；screenshots/OFF-01-offline-grep.png |
| NEI-01 | engine/eval_neighborhood.py 用 evidence_set_w4.jsonl 验证邻域扩展 | 12 条题都能对齐块；跨块题能被 ±1 邻域覆盖 | chunk_coverage_ok=12/12，cross_chunk_rows=7。严格口径 reachable_with_one_hop=5（forall，即原来的 5/7）。另计 exists_one_anchor=6：跨度 2 的五条两口径都过；w4-04 跨度 3 只过 | 部分通过 | logs/e2e_neighborhood_w4.txt；证据集说明_w4.md |

| 用例 ID | 命令 | 期望 | 实际 | 判定 | 证据 |
| --- | --- | --- | --- | --- | --- |
| | | | exists；w4-03 跨度 4 两口径都不过 | | |
| QRY-01 | `search_probe --top 5`，先 `--arm A2` 再 `--arm A4`，五条图像问句 | 文档文本塔用原问句。词表命中时 CLIP 文本塔用离线英文词。有中文 bigram 时，长度不少于 3 的拉丁串和英文词再走 trigram。不并入冻结的 25 条分数，也不改 E2E-07 | 2026-10-27，权重仍是 `(0.5, 0.3, 0.2)`。五条 A2 的金标都进 Top-5：披萨、自行车、客机和 `sample.png` 是第 1，浴室是第 2。四条场景问句 `keyword_hits=0`。A4 的 Top-5 仍是 `sample.pdf` / `sample.docx`。`sample.png` 的 A2 第 1 只说明文件被召回 | 通过 | `logs/query_rewrite_probe.txt`；检索精度基准报告.md 第 9 节 |

E2E-01 到 E2E-12 里，E2E-07 未通过，其余通过。NEI-01 的严格口径仍是 5/7，w4-03 与 w4-04 的 forall 未覆盖；存在锚点口径是 6/7。

2026-10-27 先改了查询，再把图像权重从 0.2 改为 0.49。上面的用例表和六臂表仍是权重 0.2 的记录，E2E-07 那一行保持未通过。新测量在 `logs/retrieval_bench_reweight_dev.txt`。同一句披萨问句的融合名次是 3，`000000397133.jpg` 在 Top-5 里。蓝自行车、破损淋浴墙、客机也在 Top-5 里。版面标题那条仍不在，Top-5 还是两份说明。文本和元数据十条仍全部命中。holdout 的 9/10 写在精度报告第 10 节，是第二次查看，不改上面 H2 的 5/10，也不能与 0.500 并列成提升。

## 3. 交付物

| 编号 | 文件 | 状态 |
| --- | --- | --- |
| G1 | `packages/core_storage/`、`packages/core_retrieval/` | 已完成。分析无问题，默认套件全绿。COV-01 仍是 2026-10-26 的 86.19% 与 96.34%。2026-10-27 重跑这两包：存储 637/729 = 87.38%，检索 159/165 = 96.36% |
| G2 | `packages/core_pipeline/bin/index_probe.dart`、`search_probe.dart` | 已完成。H3 通过，见 E2E-04 |
| G3 | 本报告、`reports/week4/logs/`、`screenshots/` | 已完成。E2E-07 未通过，其余用例通过。20 张终端截屏 |
| G4 | `reports/week4/检索精度基准报告.md` | 已完成。六臂、Wilson 区间、逐题表。第 2 节的 holdout 仍是一次。第 10 节的 0.900 是第二次查看 |
| G5 | `reports/week4/logs/storage_equivalence.txt` | 已完成。125/125，脚本与原始 jsonl 已对齐 |
| G6 | `docs/architecture/adr/0004-vector-storage-and-fusion.md` | 已完成。决策 1–7 各含 Alternatives/Consequences，决策 8 以「运行时」小节挂在决策 1 下并补齐这两节； |

| 编号 | 文件 | 状态 |
| --- | --- | --- |
| | | 融合后果写明不低于单通道且无正增益 |
| G7 | docs/api/storage-module-api.md、docs/api/retrieval-module-api.md | 已完成。存储文档写明 1 万块写入 444 ms、查询中位 44 ms。代次判等是 6 个字段，即决策 4 的五项，解析器名和版本分计 |
| G8 | reports/week4/coverage/*.lcov.info、覆盖率报告.md | 已完成。五个包在 2026-10-26 都在门槛之上。2026-10-27 的两包副本另档，不覆盖当天的 lcov |

reports/week4/evidence_set_w4.jsonl 有 12 条，说明在 证据集说明_w4.md。它没有并入冻结集，也没有参加打分；本轮由 engine/eval_neighborhood.py 首次用于邻域验证，结果见 NEI-01。

## 4. 证据索引

| 路径 | 对应内容 |
| --- | --- |
| reports/week4/logs/dart_analyze.txt | ANA-01 |
| reports/week4/logs/coverage.txt | UNIT-01、COV-01 的合计。2026-10-26 |
| reports/week4/logs/coverage_20261027_storage_retrieval.txt | 存储与检索的后一次覆盖率。不改 COV-01 |
| reports/week4/coverage/*.lcov.info | COV-01 的五份，加上 2026-10-27 的 core_storage_20261027.lcov.info 与 core_retrieval_20261027.lcov.info |
| reports/week4/logs/storage_equivalence.txt | H1 |
| reports/week4/logs/storage_capacity.txt | CAP-01、CAP-02 |
| reports/week4/logs/chroma_baseline.txt | CHR-01 |
| reports/week4/logs/e2e_build.txt | E2E-01 |
| reports/week4/logs/e2e_samples.txt | E2E-01 的副本 |
| reports/week4/logs/e2e_rebuild.txt | E2E-02 |
| reports/week4/logs/e2e_incremental.txt | E2E-03 |
| reports/week4/logs/e2e_query.txt | E2E-04、E2E-05、E2E-07 |
| reports/week4/logs/e2e07_clip_diagnosis.txt | E2E-07 的三条诊断。中文问句仍未通过，不并进 25 条分数 |
| reports/week4/logs/e2e_filters.txt | E2E-06、E2E-07 的 jpg 过滤 |
| reports/week4/logs/e2e_neighborhood.txt | E2E-08 |
| reports/week4/logs/e2e04_rank_after_reweight_20261027.txt | E2E-04、E2E-08、OFF-01 在权重 0.49 下的重测。617–710 第 6 名 |
| reports/week4/logs/e2e07_rerun_20261027.txt | E2E-07 在权重 0.49 下的重测。判据已满足，本行仍判未通过 |

| 路径 | 对应内容 |
| --- | --- |
| reports/week4/logs/e2e_cancel.txt | E2E-09、E2E-10 |
| reports/week4/logs/e2e_clear.txt | E2E-11、E2E-12 |
| reports/week4/logs/e2e_docs.txt | DOC-01 |
| reports/week4/logs/retrieval_bench_dev.txt | H2 的 dev |
| reports/week4/logs/retrieval_bench_holdout.txt | H2 的 holdout |
| reports/week4/logs/offline_check.txt | OFF-01 的 grep 门禁与断网说明 |
| reports/week4/logs/offline_namespace.txt | OFF-01 的断网命名空间取证（无路由、无 DNS） |
| reports/week4/logs/offline_firewall.txt | OFF-01 的出站规则。查询在规则生效时命中，规则已删除 |
| reports/week4/logs/e2e_neighborhood_w4.txt | NEI-01 的邻域验证 |
| reports/week4/logs/keyword_path_probe.txt | 关键词两条路径。不并入冻结的 25 条分数 |
| reports/week4/logs/query_rewrite_probe.txt | QRY-01。2026-10-27 查询改写，当时图像权重仍是 0.2 |
| reports/week4/logs/retrieval_bench_reweight_dev.txt | 图像权重 0.49 的 dev。不替换冻结的六臂表 |
| reports/week4/logs/retrieval_bench_reweight_holdout.txt | 图像权重 0.49 的 holdout。第二次查看，0.900 不再是无偏估计，不替换第一次的 0.500 |
| reports/week4/logs/env_w4.txt | 环境与目标文件是否存在 |
| reports/week4/检索精度基准报告.md | G4 |
| reports/week4/覆盖率报告.md | G8 |
| reports/week4/evidence_set_w4.jsonl | 邻域核对，未打分 |
| screenshots/*.png | 第 2 节有截屏的用例 |

## 5. 截屏

按第 2 节的用例 ID 排列。UNIT-01 和 DOC-01 没有单独的终端截屏，证据仍是对应日志。CAP-01 与 CAP-02 共用一张图。

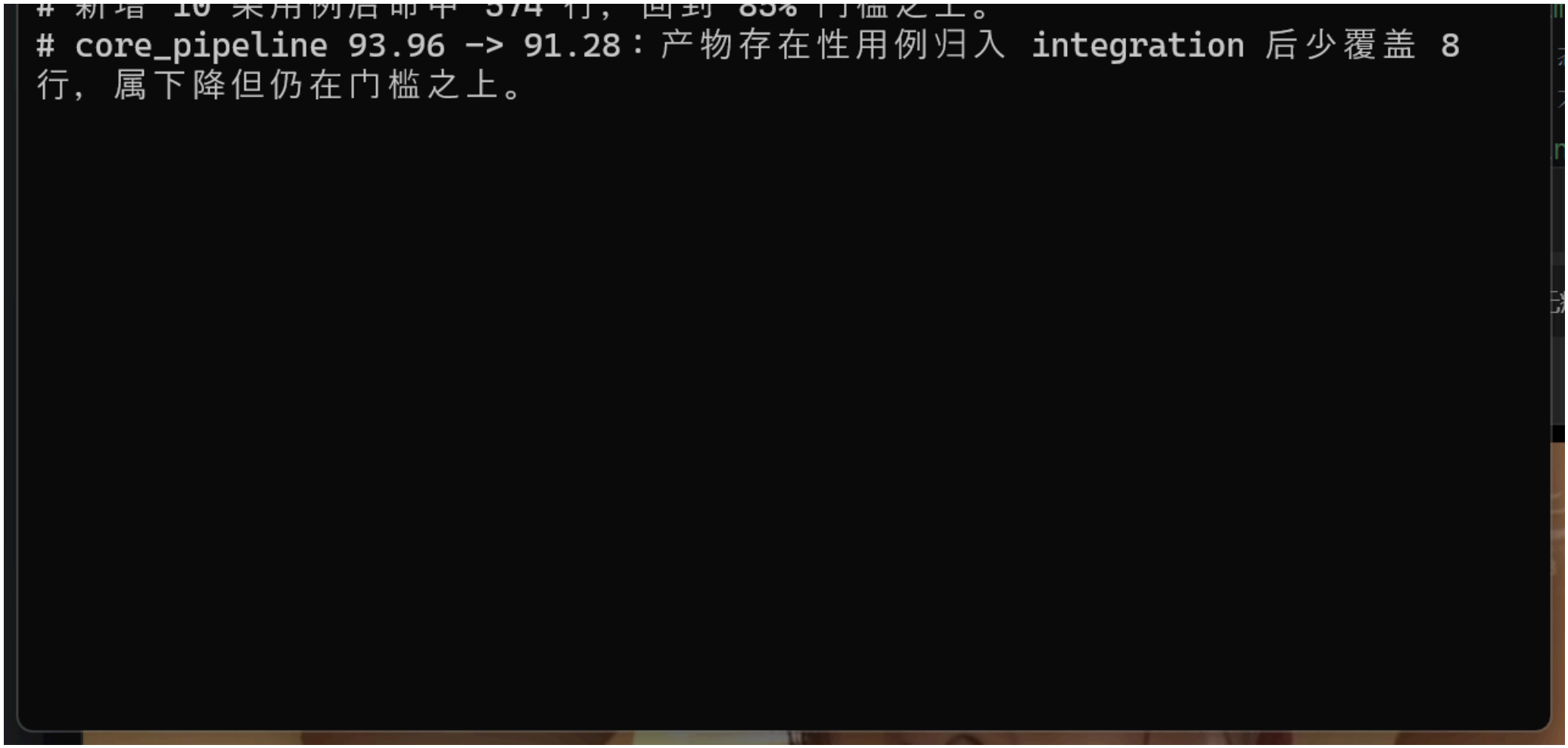
```
accessible-multimodal-retrieval on  main [✗+?] took 38s
❯ foreach ($p in 'core_storage','core_retrieval','core_pipeline') {
·     Push-Location "packages\$p"
·     dart analyze
·     Pop-Location
· }
Analyzing core_storage...
No issues found!
Analyzing core_retrieval...
No issues found!
Analyzing core_pipeline...
No issues found!
```

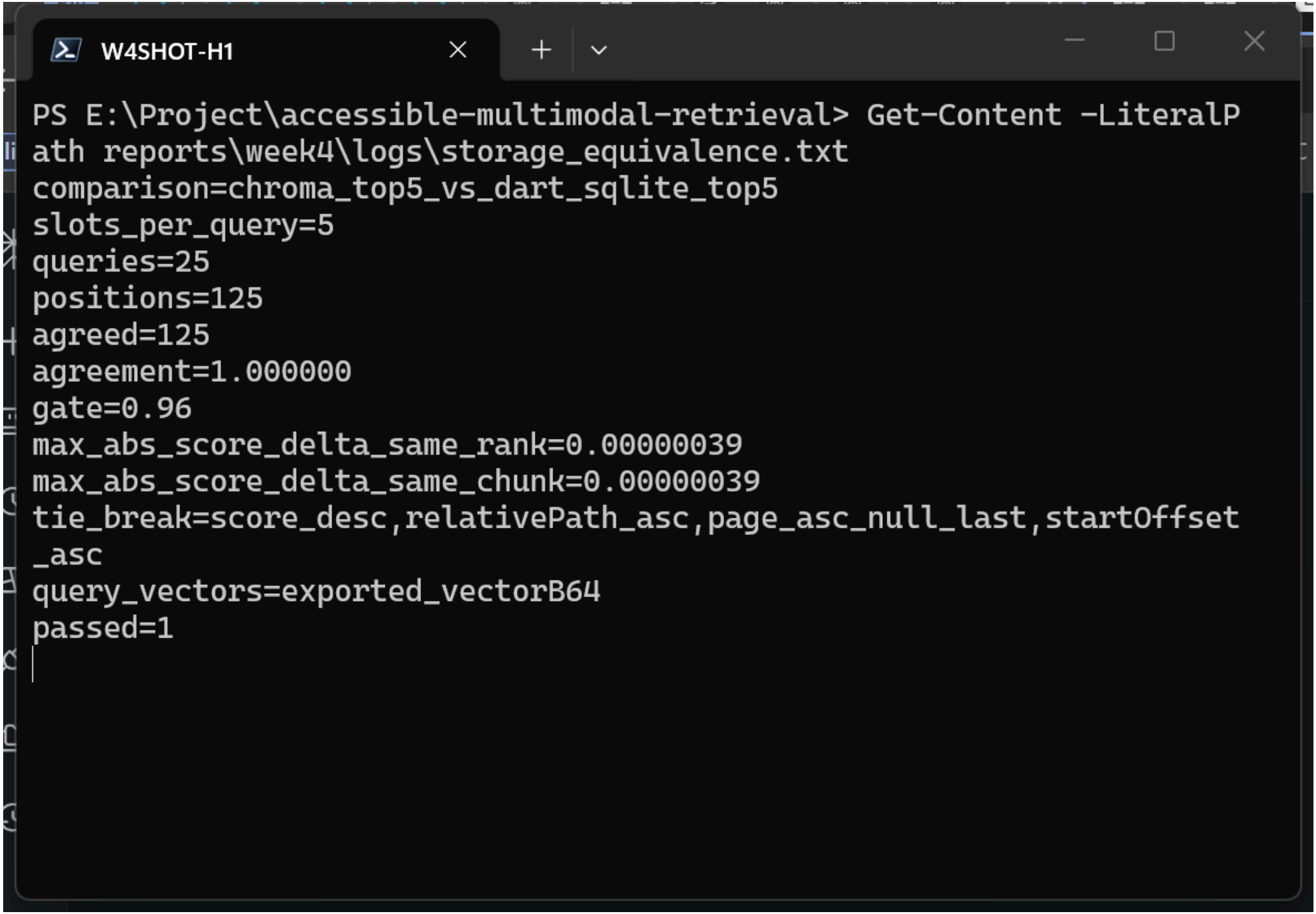
W4SHOT-COV

```
PS E:\Project\accessible-multimodal-retrieval> Get-Content -LiteralPath rep
orts\week4\logs\coverage.txt
date=2026-10-26
# From each package directory. integration tests are excluded and do not lo
ad models.
#   dart test --coverage=coverage --exclude-tags integration
#   dart run coverage:format_coverage --lcov --in=coverage --out=coverage\l
cov.info --report-on=lib --package=.
# Copies are reports/week4/coverage/<package>.lcov.info

core_storage lines_hit=574 lines_found=666 pct=86.19 suite=00:02 +23
core_retrieval lines_hit=158 lines_found=164 pct=96.34 suite=00:02 +9
core_pipeline lines_hit=272 lines_found=298 pct=91.28 suite=00:03 +23
core_parsing lines_hit=979 lines_found=1091 pct=89.73 suite=00:06 +98
core_embedding lines_hit=1250 lines_found=1458 pct=85.73 suite=00:11 +93

gate_new_packages_min=85
gate_core_parsing_min=80
gate_core_embedding_min=85
core_parsing_week2_snapshot=949/1043=90.99 on that week's 1043 instrumented
 lines
core_embedding_matches_week3_snapshot=1250/1458=85.73
# 2026-10-26: core_storage 分母 573 -> 666, 来自新增的 ScoredCandidate/rowI
dOf 与三阶段向量查询。
# 新增 10 条用例后命中 574 行，回到 85% 中枢之上
```


# 新增 10 条用例后命中 374 行，回到 85% 门槛之上。
# core_pipeline 93.96 -> 91.28：产物存在性用例归入 integration 后少覆盖 8 行，属下降但仍在门槛之上。


W4SHOT-H1
PS E:\Project\accessible-multimodal-retrieval> Get-Content -LiteralP
ath reports\week4\logs\storage_equivalence.txt
comparison=chroma_top5_vs_dart_sqlite_top5
slots_per_query=5
queries=25
positions=125
agreed=125
agreement=1.000000
gate=0.96
max_abs_score_delta_same_rank=0.00000039
max_abs_score_delta_same_chunk=0.00000039
tie_break=score_desc,relativePath_asc,page_asc_null_last,startOffset
_asc
query_vectors=exported_vectorB64
passed=1

W4SHOT-CAP

```
PS E:\Project\accessible-multimodal-retrieval> Get-Content -LiteralP
ath reports\week4\logs\storage_capacity.txt
sqlite_version=3.53.4
journal_mode=wal
n=1500 write_ms=95 query_ms_median=10 query_ms_min=8 query_ms_max=18
n=10000 write_ms=444 query_ms_median=44 query_ms_min=41 query_ms_max
=50
explain_filter=extension_in_sql
explain=3|0|60|SEARCH chunk USING INDEX idx_chunk_space (space=? AND
 dim=?)
write_limit_ms=600
query_limit_ms=60
```

```
accessible-multimodal-retrieval on  main [✗+?]
❯ Get-Content -LiteralPath reports\week4\logs\chroma_baseline.txt
chroma_vs_numpy_positions=125
chroma_vs_numpy_agreed=125
chroma_vs_numpy_agreement=1.000000
score=1-distance
hnsw_space=cosine
chromadb_version=1.5.9
queries=25
hit@5=0.880000 [0.700442, 0.958332] (22/25)
mrr@10=0.795714
metric_layer=storage_baseline
note=query vectors are the exported question embeddings; this is the Chroma control, not the product retriever
```

```
accessible-multimodal-retrieval on  main [✗+?]
❯ Get-Content -LiteralPath reports\week4\logs\e2e_build.txt
progress 33/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 34/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 35/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 36/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 37/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 38/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 39/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 40/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 41/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 42/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 43/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 44/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 45/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 46/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 47/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 48/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 49/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 50/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
path=E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
key_prefix=datasets/coco/val_sample/images
ingest_total=50 ingest_ok=50 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=1418
documents=50 written=50 reused=0 failed=0 cancelled=false chunks=50 index_ms=7048
stats_chunks=147 stats_documents=53 stats_document_text=96 stats_clip_shared=51 stats_keyword_terms=0
index_ms_total=19093
```

```
accessible-multimodal-retrieval on  main [✗+?]
❯ dart run packages/core_pipeline/bin/index_probe.dart --db reports/week4/.index/evidence.sqlite --path datasets/samples --path datasets/coco/val_sample/images
progress 33/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 34/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 35/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 36/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 37/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 38/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 39/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 40/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 41/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 42/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 43/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 44/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 45/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 46/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 47/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 48/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 49/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
progress 50/50 E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
path=E:\Project\accessible-multimodal-retrieval\datasets/coco/val_sample/images
key_prefix=datasets/coco/val_sample/images
ingest_total=50 ingest_ok=50 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=1298
documents=50 written=0 reused=50 failed=0 cancelled=false chunks=0 index_ms=1316
stats_chunks=147 stats_documents=53 stats_document_text=96 stats_clip_shared=51 stats_keyword_terms=0
index_ms_total=7465
```

```
accessible-multimodal-retrieval on  main [✗+?] took 23s
❯ $inc = Join-Path $env:TEMP 'w4-shot-incremental'
· if (Test-Path $inc) { Remove-Item -LiteralPath $inc -Recurse -Force }
· New-Item -ItemType Directory -Path $inc | Out-Null
· Copy-Item docs\architecture\adr\0001-parsing-runtime-dart-package.md (Join-Path $inc 'a.md') …
progress 0/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 1/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 2/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 3/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
path=C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
key_prefix=C:/Users/11758/AppData/Local/Temp/w4-shot-incremental
ingest_total=3 ingest_ok=3 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=80
documents=3 written=3 reused=0 failed=0 cancelled=false chunks=393 index_ms=17382
stats_chunks=393 stats_documents=3 stats_document_text=393 stats_clip_shared=0 stats_keyword_terms=3854
index_ms_total=17382
db=C:\Users\11758\AppData\Local\Temp\w4-shot-incremental\inc.sqlite
progress 0/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 1/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 2/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
progress 3/3 C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
path=C:\Users\11758\AppData\Local\Temp\w4-shot-incremental
key_prefix=C:/Users/11758/AppData/Local/Temp/w4-shot-incremental
ingest_total=3 ingest_ok=3 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=41
documents=3 written=1 reused=2 failed=0 cancelled=false chunks=85 index_ms=5158
stats_chunks=394 stats_documents=3 stats_document_text=394 stats_clip_shared=0 stats_keyword_terms=3871
index_ms_total=5158
```

```
accessible-multimodal-retrieval on  main [✗?] took 38s
❯ dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 10 --query "工具靠什么判断文件类型？"
query=工具靠什么判断文件类型？
arm=A5 expand=true top=10 encode_ms=2237 store_ms=35 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=3 score=0.007937 key=datasets/samples/sample.docx file=sample.docx page=null start=306 end=426 channels=documentText modality=text
snippet=Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different file types (such as PPT, X\nLS, and PDF).
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=617 end=710 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=5 score=0.007692 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=626 end=719 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=6 score=0.007576 key=datasets/samples/sample.docx file=sample.docx page=null start=440 end=560 channels=documentText modality=text
snippet=LS, and PDF).\nAll of these file types can be parsed through a single interface, making Tika useful for search engine indexing, content\nanalysis, translation, and much more.
rank=7 score=0.007463 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=326 end=419 channels=documentText modality=text
snippet=but more work remains. Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
rank=8 score=0.007353 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=332 end=425 channels=documentText modality=text
snippet=but more work remains. Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
rank=9 score=0.007246 key=datasets/samples/sample.docx file=sample.docx page=null start=599 end=709 channels=documentText modality=text
snippet=analysis, translation, and much more.\nTika identifies a file’s type from its bytes, so a missing or misleading extension does not change the result.\nWhere to start
rank=10 score=0.007143 key=datasets/samples/sample.docx file=sample.docx page=null start=725 end=826 channels=documentText modality=text
snippet=Where to start\nSupported Formats — what Tika parses, family by family, plus the generated parser-to-media-type list.\nUsing Tika — choose an integration method: Java API, command line, REST server, or gRPC.
```

```
accessible-multimodal-retrieval on  main [✗?] took 9s
❯ dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 10 --query "这份说明在能够被完全信任之前，还要求人做什么？"
query=这份说明在能够被完全信任之前，还要求人做什么？
arm=A5 expand=true top=10 encode_ms=2584 store_ms=37 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=160 end=253 channels=documentText modality=text
snippet=This reference guide was generated with the assistance of AI and requires human review\nbefore it can be fully trusted. This documentation serves as an example and a starting point,\nbut more work remains. Contributions and corrections are welcome.
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=164 end=257 channels=documentText modality=text
snippet=This reference guide was generated with the assistance of AI and requires human review\nbefore it can be fully trusted. This documentation serves as an example and a starting point,\nbut more work remains. Contributions and corrections are welcome.
rank=3 score=0.007937 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=1232 end=1314 channels=documentText modality=text
snippet=content you do not control.\n• Security — the trust model, and what to lock down before exposing tika-server or\ntika-grpc.
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=1252 end=1334 channels=documentText modality=text
snippet=content you do not control.\n• Security — the trust model, and what to lock down before exposing tika-server or\ntika-grpc.
rank=5 score=0.007692 key=datasets/samples/sample.docx file=sample.docx page=null start=1160 end=1251 channels=documentText modality=text
snippet=Setting Limits and Robustness — bounding time, memory and output when parsing content you do not control.\nSecurity — the trust model, and what to lock down before exposing tika-server or tika-grpc.\nMigrating to 4.x — what changed since 3.x and what you have to update.
rank=6 score=0.007576 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=2005 end=2053 channels=documentText modality=text
snippet=https://tika.apache.org/docs/4.0.x/advanced/robustness.html\nhttps://tika.apache.org/docs/4.0.x/security.html\nhttps://tika.apache.org/docs/4.0.x/migration-to-4x/index.html
rank=7 score=0.007463 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=2041 end=2089 channels=documentText modality=text
snippet=https://tika.apache.org/docs/4.0.x/advanced/robustness.html\nhttps://tika.apache.org/docs/4.0.x/security.html\nhttps://tika.apache.org/docs/4.0.x/migration-to-4x/index.html
rank=8 score=0.007353 key=datasets/samples/sample.docx file=sample.docx page=null start=59 end=177 channels=documentText modality=text
snippet=Where to start\nThis reference guide was generated with the assistance of AI and requires human review before it can be fully trusted.\nThis documentation serves as an example and a starting point, but more work remains.
rank=9 score=0.007246 key=datasets/samples/sample.docx file=sample.docx page=null start=178 end=262 channels=documentText modality=text
snippet=This reference guide was generated with the assistance of AI and requires human review before it can be fully trusted.\nThis documentation serves as an example and a starting point, but more work remains.\nContributions and corrections are welcome.
rank=10 score=0.007143 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=1881 end=1944 channels=documentText modality=text
snippet=https://tika.apache.org/docs/4.0.x/configuration/index.html\nhttps://tika.apache.org/docs/4.0.x/advanced/setting-limits.html\nhttps://tika.apache.org/docs/4.0.x/advanced/robustness.html
```

```
accessible-multimodal-retrieval on  main [✗+?] took 11s
❯ dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 5 --extension .pdf --query "工具靠什么判断文件类型？"
· dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 5 --path-prefix sample.pdf --query "工具靠什么判断文件类型？"
· dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 3 --modified-after 2020-01-01T00:00:00Z --query "工具靠什么判断文件类型？"
· dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 3 --modified-after 2099-01-01T00:00:00Z --query "工具靠什么判断文件类型？"
query=工具靠什么判断文件类型？
arm=A5 expand=true top=5 encode_ms=2379 store_ms=30 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation, and
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation, and
rank=3 score=0.007937 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=617 end=710 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=626 end=719 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=5 score=0.007692 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=326 end=419 channels=documentText modality=text
snippet=but more work remains. Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
query=工具靠什么判断文件类型？
arm=A5 expand=true top=5 encode_ms=2301 store_ms=34 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation, and
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation, and
rank=3 score=0.007937 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=617 end=710 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=626 end=719 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=5 score=0.007692 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=326 end=419 channels=documentText modality=text
snippet=but more work remains. Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
query=工具靠什么判断文件类型？
arm=A5 expand=true top=3 encode_ms=2138 store_ms=35 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation, and
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation, and
rank=3 score=0.007937 key=datasets/samples/sample.docx file=sample.docx page=null start=306 end=426 channels=documentText modality=text
snippet=Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different file types (such as PPT, X\nLS, and PDF).
query=工具靠什么判断文件类型？
arm=A5 expand=true top=0 encode_ms=2408 store_ms=11 fuse_ms=0
hits=0
```

```
accessible-multimodal-retrieval on  main [✗+?] took 38s
❯ dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 5 --extension .jpg --query "哪张图里有人在厨房做披萨？"
query=哪张图里有人在厨房做披萨？
arm=A5 expand=true top=5 encode_ms=2575 store_ms=33 fuse_ms=3
rank=1 score=0.003279 key=datasets/coco/val_sample/images/000000403385.jpg file=000000403385.jpg page=null start=0 end=0 channels=clipShared modality=image
snippet=
rank=2 score=0.003226 key=datasets/coco/val_sample/images/000000296649.jpg file=000000296649.jpg page=null start=0 end=0 channels=clipShared modality=image
snippet=
rank=3 score=0.003175 key=datasets/coco/val_sample/images/000000463730.jpg file=000000463730.jpg page=null start=0 end=0 channels=clipShared modality=image
snippet=
rank=4 score=0.003125 key=datasets/coco/val_sample/images/000000006818.jpg file=000000006818.jpg page=null start=0 end=0 channels=clipShared modality=image
snippet=
rank=5 score=0.003077 key=datasets/coco/val_sample/images/000000329323.jpg file=000000329323.jpg page=null start=0 end=0 channels=clipShared modality=image
snippet=
```

```
accessible-multimodal-retrieval on  main [✗+?] took 9s
❯ dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 5 --arm A5 --query "工具靠什么判断文件类型？"
· dart run packages/core_pipeline/bin/search_probe.dart --db reports/week4/.index/evidence.sqlite --top 5 --arm A4 --query "工具靠什么判断文件类型？"
query=工具靠什么判断文件类型？
arm=A5 expand=true top=5 encode_ms=2614 store_ms=36 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different\nfile types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single\ninterface, making Tika useful for search engine indexing, content analysis, translation,
and
rank=3 score=0.007937 key=datasets/samples/sample.docx file=sample.docx page=null start=306 end=426 channels=documentText modality=text
snippet=Contributions and corrections are welcome.\nThe Apache Tika toolkit detects and extracts metadata and text from over a thousand different file types (such as PPT, X\nLS, and PDF).
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=617 end=710 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
rank=5 score=0.007692 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=626 end=719 channels=documentText modality=text
snippet=interface, making Tika useful for search engine indexing, content analysis, translation, and\nmuch more. Tika identifies a file’s type from its bytes, so a missing or misleading extension\ndoes not change the result.
query=工具靠什么判断文件类型？
arm=A4 expand=false top=5 encode_ms=2806 store_ms=22 fuse_ms=3
rank=1 score=0.008197 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=423 end=517 channels=documentText modality=text
snippet=file types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
rank=2 score=0.008065 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=430 end=524 channels=documentText modality=text
snippet=file types (such as PPT, XLS, and PDF). All of these file types can be parsed through a single
rank=3 score=0.007937 key=datasets/samples/sample.docx file=sample.docx page=null start=306 end=426 channels=documentText modality=text
snippet=The Apache Tika toolkit detects and extracts metadata and text from over a thousand different file types (such as PPT, X
rank=4 score=0.007813 key=datasets/samples/sample.pdf file=sample.pdf page=1 start=617 end=710 channels=documentText modality=text
snippet=much more. Tika identifies a file’s type from its bytes, so a missing or misleading extension
rank=5 score=0.007692 key=datasets/samples/sample.pdf file=sample.pdf page=2 start=626 end=719 channels=documentText modality=text
snippet=much more. Tika identifies a file’s type from its bytes, so a missing or misleading extension
```

```
accessible-multimodal-retrieval on  main [✗+?] took 22s
❯ $cancel = Join-Path $env:TEMP 'w4-shot-cancel'
· if (Test-Path $cancel) { Remove-Item -LiteralPath $cancel -Recurse -Force }
· New-Item -ItemType Directory -Path $cancel | Out-Null
· Set-Content -LiteralPath (Join-Path $cancel 'a.txt') -Value '第一份文件保留。离线检索可以没有网络。' -Encoding utf8
· Set-Content -LiteralPath (Join-Path $cancel 'b.txt') -Value '第二份文件不应写入。键盘和显示器。' -Encoding utf8
· Set-Content -LiteralPath (Join-Path $cancel 'c.txt') -Value '第三份文件不应写入。向量数据库。' -Encoding utf8
· dart run packages/core_pipeline/bin/index_probe.dart --db (Join-Path $cancel 'cancel.sqlite') --path $cancel --cancel-after 1
· dart run packages/core_pipeline/bin/search_probe.dart --db (Join-Path $cancel 'cancel.sqlite') --top 5 --query "离线检索可以没有网络"
db=C:\Users\11758\AppData\Local\Temp\w4-shot-cancel\cancel.sqlite
progress 0/3 C:\Users\11758\AppData\Local\Temp\w4-shot-cancel
progress 1/3 C:\Users\11758\AppData\Local\Temp\w4-shot-cancel
path=C:\Users\11758\AppData\Local\Temp\w4-shot-cancel
key_prefix=C:/Users/11758/AppData/Local/Temp/w4-shot-cancel
ingest_total=3 ingest_ok=3 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=99
documents=3 written=1 reused=0 failed=0 cancelled=true chunks=1 index_ms=3070
stats_chunks=1 stats_documents=1 stats_document_text=1 stats_clip_shared=0 stats_keyword_terms=15
index_ms_total=3070
query=离线检索可以没有网络
arm=A5 expand=true top=1 encode_ms=2411 store_ms=18 fuse_ms=2
rank=1 score=0.013115 key=C:/Users/11758/AppData/Local/Temp/w4-shot-cancel/a.txt file=a.txt page=null start=0 end=19 channels=documentText,keyword modality=text
snippet=第一份文件保留。离线检索可以没有网络。
```

```
accessible-multimodal-retrieval on  main [✗+?] took 32s
❯ $bad = Join-Path $env:TEMP 'w4-shot-bad'
· if (Test-Path $bad) { Remove-Item -LiteralPath $bad -Recurse -Force }
· New-Item -ItemType Directory -Path $bad | Out-Null
· Set-Content -LiteralPath (Join-Path $bad 'good.txt') -Value '完好的文本文件仍应入库。' -Encoding utf8
· Set-Content -LiteralPath (Join-Path $bad 'bad.docx') -Value 'this is not a docx archive' -Encoding utf8
· dart run packages/core_pipeline/bin/index_probe.dart --db (Join-Path $bad 'bad.sqlite') --path $bad
· dart run packages/core_pipeline/bin/search_probe.dart --db (Join-Path $bad 'bad.sqlite') --top 5 --query "完好的文本文件仍应入库"
db=C:\Users\11758\AppData\Local\Temp\w4-shot-bad\bad.sqlite
progress 0/1 C:\Users\11758\AppData\Local\Temp\w4-shot-bad
progress 1/1 C:\Users\11758\AppData\Local\Temp\w4-shot-bad
path=C:\Users\11758\AppData\Local\Temp\w4-shot-bad
key_prefix=C:/Users/11758/AppData/Local/Temp/w4-shot-bad
ingest_total=2 ingest_ok=1 ingest_partial=0 ingest_skipped=0 ingest_failed=1 ingest_cancelled=false ingest_ms=330
documents=1 written=1 reused=0 failed=0 cancelled=false chunks=1 index_ms=3054
stats_chunks=1 stats_documents=1 stats_document_text=1 stats_clip_shared=0 stats_keyword_terms=10
index_ms_total=3054
query=完好的文本文件仍应入库
arm=A5 expand=true top=1 encode_ms=2564 store_ms=47 fuse_ms=15
rank=1 score=0.013115 key=C:/Users/11758/AppData/Local/Temp/w4-shot-bad/good.txt file=good.txt page=null start=0 end=12 channels=documentText,keyword modality=text
snippet=完好的文本文件仍应入库。
```

```
accessible-multimodal-retrieval on  main [✗+?] took 30s
❯ $empty = Join-Path $env:TEMP 'w4-shot-empty'
· if (Test-Path $empty) { Remove-Item -LiteralPath $empty -Recurse -Force }
· New-Item -ItemType Directory -Path $empty | Out-Null
· dart run packages/core_pipeline/bin/index_probe.dart --db (Join-Path $empty 'empty.sqlite') --clear
· dart run packages/core_pipeline/bin/search_probe.dart --db (Join-Path $empty 'empty.sqlite') --top 5 --query "离线检索"
db=C:\Users\11758\AppData\Local\Temp\w4-shot-empty\empty.sqlite
clear_before_chunks=0 clear_after_chunks=0
query=离线检索
arm=A5 expand=true top=0 encode_ms=1637 store_ms=10 fuse_ms=0
hits=0
```

```
accessible-multimodal-retrieval on  main [X+?] took 18s
❯ $empty = Join-Path $env:TEMP 'w4-shot-empty'
· Set-Content -LiteralPath (Join-Path $empty 'note.txt') -Value '清库前的句子。离线检索。' -Encoding utf8
· dart run packages/core_pipeline/bin/index_probe.dart --db (Join-Path $empty 'empty.sqlite') --path $empty
· dart run packages/core_pipeline/bin/search_probe.dart --db (Join-Path $empty 'empty.sqlite') --top 5 --query "清库前的句子"
· dart run packages/core_pipeline/bin/index_probe.dart --db (Join-Path $empty 'empty.sqlite') --clear
· dart run packages/core_pipeline/bin/search_probe.dart --db (Join-Path $empty 'empty.sqlite') --top 5 --query "清库前的句子"
db=C:\Users\11758\AppData\Local\Temp\w4-shot-empty\empty.sqlite
progress 0/1 C:\Users\11758\AppData\Local\Temp\w4-shot-empty
progress 1/1 C:\Users\11758\AppData\Local\Temp\w4-shot-empty
path=C:\Users\11758\AppData\Local\Temp\w4-shot-empty
key_prefix=C:/Users/11758/AppData/Local/Temp/w4-shot-empty
ingest_total=1 ingest_ok=1 ingest_partial=0 ingest_skipped=0 ingest_failed=0 ingest_cancelled=false ingest_ms=86
documents=1 written=1 reused=0 failed=0 cancelled=false chunks=1 index_ms=2163
stats_chunks=1 stats_documents=1 stats_document_text=1 stats_clip_shared=0 stats_keyword_terms=8
index_ms_total=2163
query=清库前的句子
arm=A5 expand=true top=1 encode_ms=1898 store_ms=17 fuse_ms=1
rank=1 score=0.013115 key=C:/Users/11758/AppData/Local/Temp/w4-shot-empty/note.txt file=note.txt page=null start=0 end=12 channels=documentText,keyword modality=text
snippet=清库前的句子。离线检索。
db=C:\Users\11758\AppData\Local\Temp\w4-shot-empty\empty.sqlite
clear_before_chunks=1 clear_after_chunks=0
query=清库前的句子
arm=A5 expand=true top=0 encode_ms=1775 store_ms=8 fuse_ms=0
hits=0
```

```
accessible-multimodal-retrieval on  main [X+?]
❯ Select-String -LiteralPath reports\week4\logs\retrieval_bench_dev.txt -Pattern '^(parameters_changed_after_freeze|weights|k=|holdout_scored_runs|arm=)'

reports\week4\logs\retrieval_bench_dev.txt:1:parameters_changed_after_freeze=0
reports\week4\logs\retrieval_bench_dev.txt:2:weights=text:0.5,keyword:0.3,image:0.2
reports\week4\logs\retrieval_bench_dev.txt:3:k=60
reports\week4\logs\retrieval_bench_dev.txt:4:holdout_scored_runs=0
reports\week4\logs\retrieval_bench_dev.txt:20:arm=A0 split=dev n=15 multi_evidence=8 hit@5=0.000000 [0.000000, 0.203883] (0/15) mrr@10=0.000000 recall@5=0.000000 recall@10=0.000000 encode_ms_first=0 encode_ms_median_after_first=0.0 store_ms_median=0.0 fuse_ms_median=0.0
reports\week4\logs\retrieval_bench_dev.txt:36:arm=A1 split=dev n=15 multi_evidence=8 hit@5=0.666667 [0.417135, 0.848237] (10/15) mrr@10=0.622222 recall@5=0.633333 recall@10=0.666667 encode_ms_first=1732 encode_ms_median_after_first=79.0 store_ms_median=1.0 fuse_ms_median=0.0
reports\week4\logs\retrieval_bench_dev.txt:52:arm=A2 split=dev n=15 multi_evidence=8 hit@5=0.133333 [0.037361, 0.378820] (2/15) mrr@10=0.096296 recall@5=0.133333 recall@10=0.200000 encode_ms_first=437 encode_ms_median_after_first=52.5 store_ms_median=1.0 fuse_ms_median=0.0
reports\week4\logs\retrieval_bench_dev.txt:68:arm=A3 split=dev n=15 multi_evidence=8 hit@5=0.666667 [0.417135, 0.848237] (10/15) mrr@10=0.622222 recall@5=0.633333 recall@10=0.666667 encode_ms_first=37 encode_ms_median_after_first=83.5 store_ms_median=2.0 fuse_ms_median=0.0
reports\week4\logs\retrieval_bench_dev.txt:84:arm=A4 split=dev n=15 multi_evidence=8 hit@5=0.666667 [0.417135, 0.848237] (10/15) mrr@10=0.622222 recall@5=0.633333 recall@10=0.666667 encode_ms_first=67 encode_ms_median_after_first=78.5 store_ms_median=4.0 fuse_ms_median=0.0
reports\week4\logs\retrieval_bench_dev.txt:100:arm=A5 split=dev n=15 multi_evidence=8 hit@5=0.666667 [0.417135, 0.848237] (10/15) mrr@10=0.622222 recall@5=0.633333 recall@10=0.666667 encode_ms_first=59 encode_ms_median_after_first=127.0 store_ms_median=5.0 fuse_ms_median=0.0
```

```
accessible-multimodal-retrieval on  main [X+?]
❯ Get-Content -LiteralPath reports\week4\logs\retrieval_bench_holdout.txt
parameters_changed_after_freeze=0
weights=text:0.5,keyword:0.3,image:0.2
k=60
holdout_scored_runs=1
ev-16 split=holdout arm=A4 hit@5=1 rank=1 channels=documentText top5=[datasets/samples/sample.docx#pna@1, datasets/samples/sample.pdf#p1@2, datasets/samples/sample.pdf#p2@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
ev-17 split=holdout arm=A4 hit@5=1 rank=1 channels=documentText top5=[datasets/samples/sample.pdf#p1@1, datasets/samples/sample.pdf#p2@2, datasets/samples/sample.docx#pna@3, datasets/samples/sample.docx#pna@4, datasets/samples/sample.pdf#p1@5]
ev-18 split=holdout arm=A4 hit@5=1 rank=1 channels=documentText top5=[datasets/samples/sample.docx#pna@1, datasets/samples/sample.pdf#p1@2, datasets/samples/sample.pdf#p2@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
ev-19 split=holdout arm=A4 hit@5=1 rank=1 channels=documentText top5=[datasets/samples/sample.docx#pna@1, datasets/samples/sample.pdf#p1@2, datasets/samples/sample.pdf#p2@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
ev-20 split=holdout arm=A4 hit@5=1 rank=1 channels=documentText top5=[datasets/samples/sample.docx#pna@1, datasets/samples/sample.pdf#p1@2, datasets/samples/sample.pdf#p2@3, datasets/samples/sample.docx#pna@4, datasets/samples/sample.pdf#p1@5]
ev-21 split=holdout arm=A4 hit@5=0 rank=0 channels= top5=[datasets/samples/sample.pdf#p1@1, datasets/samples/sample.pdf#p2@2, datasets/samples/sample.docx#pna@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
ev-22 split=holdout arm=A4 hit@5=0 rank=0 channels= top5=[datasets/samples/sample.pdf#p1@1, datasets/samples/sample.pdf#p2@2, datasets/samples/sample.pdf#p1@3, datasets/samples/sample.pdf#p2@4, datasets/samples/sample.pdf#p1@5]
ev-23 split=holdout arm=A4 hit@5=0 rank=0 channels= top5=[datasets/samples/sample.pdf#p1@1, datasets/samples/sample.pdf#p2@2, datasets/samples/sample.pdf#p1@3, datasets/samples/sample.pdf#p2@4, datasets/samples/sample.pdf#p1@5]
ev-24 split=holdout arm=A4 hit@5=0 rank=0 channels= top5=[datasets/samples/sample.pdf#p1@1, datasets/samples/sample.pdf#p2@2, datasets/samples/sample.docx#pna@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
ev-25 split=holdout arm=A4 hit@5=0 rank=0 channels= top5=[datasets/samples/sample.docx#pna@1, datasets/samples/sample.pdf#p1@2, datasets/samples/sample.pdf#p2@3, datasets/samples/sample.pdf#p1@4, datasets/samples/sample.pdf#p2@5]
arm=A4 split=holdout n=10 multi_evidence=4 hit@5=0.500000 [0.236593, 0.763407] (5/10) mrr@10=0.500000 recall@5=0.500000 recall@10=0.500000 encode_ms_first=2134 encode_ms_median_after_first=55.0 store_ms_median=21.5 fuse_ms_median=0.0
```

```
accessible-multimodal-retrieval on  main [✗+?]
❯ Get-Content -LiteralPath reports\week4\logs\offline_check.txt
# offline grep 2026-10-25T16:09:01.4371110+08:00
# roots: packages/core_storage/lib packages/core_retrieval/lib packages/core_pipeline/lib
# pattern: HttpClient|Socket|import .http|package:http
# files_scanned=40
# matches=0
# network_disabled=0
# note: grep gate only; the machine network was left enabled
```

# 检索精度基准报告

## 1. 口径

证据 `id` 等于 `documentKey`，或证据路径以 `/` 加相对路径结尾。Hit@5 是 Top-5 里出现任一相关文件。MRR@10 用第一次命中的名次。多文件题同时报 Recall。比例写成 `0.666667 [0.417135, 0.848237] (10/15)`。区间是 Wilson 95%。

第 2 节到第 8 节的权重是（文本，关键词，图像）= `(0.5, 0.3, 0.2)`，`k=60`。关掉一路不是改权重。两份冻结日志的第一行都是 `parameters_changed_after_freeze=0`。dev 先跑完全部六臂，之后 holdout 只跑 A4 一次。`retrieval_bench_dev.txt` 写着 `holdout_scored_runs=0`，`retrieval_bench_holdout.txt` 写着 `holdout_scored_runs=1`。2026-10-27 把图像权重改为 0.49，`parameters_changed_after_freeze=1`。那次测量在第 10 节，不改第 2 节。

## 2. 六臂

dev 15 条，其中 8 条有多份证据。holdout 10 条，其中 4 条有多份证据。A1、A3、A4、A5 在 15 条 dev 上的 Hit@5、名次和 Top-5 逐题相同。

| 臂 | 配置 | dev Hit@5 | dev MRR@10 | dev Recall@5 | dev Recall@10 | holdout Hit@5 | holdout MRR@10 |
|---|---|---|---|---|---|---|---|
| A0 | 只开关键词，不扩展邻域 | 0.000000 [0.000000, 0.203883] (0/15) | 0.000000 | 0.000000 | 0.000000 | 未跑 | 未跑 |
| A1 | 只开 `documentText` | 0.666667 [0.417135, 0.848237] (10/15) | 0.622222 | 0.633333 | 0.666667 | 未跑 | 未跑 |

| 臂 | 配置 | dev Hit@5 | dev MRR@10 | dev Recall@5 | dev Recall@10 | holdout Hit@5 | holdout MRR@10 |
|---|---|---|---|---|---|---|---|
| A2 | 只开 `clipShared` | 0.133333 [0.037361, 0.378820] (2/15) | 0.096296 | 0.133333 | 0.200000 | 未跑 | 未跑 |
| A3 | 文本 + 关键词 | 0.666667 [0.417135, 0.848237] (10/15) | 0.622222 | 0.633333 | 0.666667 | 未跑 | 未跑 |
| A4 | 三路，不扩展邻域 | 0.666667 [0.417135, 0.848237] (10/15) | 0.622222 | 0.633333 | 0.666667 | 0.500000 [0.236593, 0.763407] (5/10) | 0.500000 |
| A5 | 三路 + 邻域，产品默认 | 0.666667 [0.417135, 0.848237] (10/15) | 0.622222 | 0.633333 | 0.666667 | 未跑 | 未跑 |

holdout 的 Recall@5 和 Recall@10 都是 0.500000。邻域不改名次，所以 A5 和 A4 的 Hit@5、MRR 相同。

命令：

```
engine\.venv\Scripts\python.exe engine\eval_retrieval.py --arms A0,A1,A2,A3,A4,A5 --split dev
engine\.venv\Scripts\python.exe engine\eval_retrieval.py --arms A4 --split holdout
```

## 3. 逐题

A4 是融合臂。A2 列只在 dev 上有数。名次 0 表示 Top-10 里没有相关文件。

| 题 | 划分 | 模态 | A4 Hit@5 | A4 名次 | A2 名次 | 说明 |
|---|---|---|---|---|---|---|
| ev-01 | dev | 文本 | 1 | 1 | 0 | 文件命中 `sample.pdf` |
| ev-02 | dev | 文本 | 1 | 1 | 0 | 文件命中 `sample.docx` |
| ev-03 | dev | 文本 | 1 | 1 | 0 | 文件命中 `sample.pdf` |
| ev-04 | dev | 文本 | 1 | 1 | 0 | 文件命中 `sample.docx` |
| ev-05 | dev | 文本 | 1 | 1 | 0 | 文件命中 `sample.pdf` |
| ev-06 | dev | 文本 | 1 | 1 | 0 | 文件命中 `sample.pdf` |
| ev-07 | dev | 文本 | 1 | 1 | 0 | 文件命中 `sample.docx` |
| ev-08 | dev | 文本 | 1 | 1 | 0 | 跨页题是两页重复排版，文件命中 |

| 题 | 划分 | 模态 | A4 Hit@5 | A4 名次 | A2 名次 | 说明 |
|---|---|---|---|---|---|---|
| ev-09 | dev | 元数据 | 1 | 3 | 0 | 创建者在 docProps，不在正文。Top-5 有 `sample.docx` |
| ev-10 | dev | 元数据 | 1 | 1 | 0 | 标题在 docProps。文件命中，不是属性跨度命中 |
| ev-11 | dev | 图像 | 0 | 0 | 0 | 披萨图 `000000397133.jpg`。中文问句只开图像也不在 Top-5。英文证据句另见诊断日志，不改本行分数 |
| ev-12 | dev | 图像 | 0 | 0 | 0 | 蓝自行车。只开图像也不在 Top-5 |
| ev-13 | dev | 图像 | 0 | 0 | 1 | 破损淋浴墙。只开图像是第 1 名，融合后掉出 |
| ev-14 | dev | 图像 | 0 | 0 | 9 | 客机。只开图像排第 9，未进 Hit@5 |
| ev-15 | dev | 图像 | 0 | 0 | 3 | `sample.png` 标题。只开图像是第 3 名，融合后掉出 |
| ev-16 | holdout | 文本 | 1 | 1 | 未跑 | 文件命中 |
| ev-17 | holdout | 文本 | 1 | 1 | 未跑 | 跨页重复排版，文件命中 |
| ev-18 | holdout | 文本 | 1 | 1 | 未跑 | 文件命中 |
| ev-19 | holdout | 文本 | 1 | 1 | 未跑 | 文件命中 |
| ev-20 | holdout | 元数据 | 1 | 1 | 未跑 | 创建时间在 docProps。文件命中 |
| ev-21 | holdout | 图像 | 0 | 0 | 未跑 | 融合 Top-5 全是文本文档 |
| ev-22 | holdout | 图像 | 0 | 0 | 未跑 | 同上 |
| ev-23 | holdout | 图像 | 0 | 0 | 未跑 | 同上 |

| 题 | 划分 | 模态 | A4 Hit@5 | A4 名次 | A2 名次 | 说明 |
|---|---|---|---|---|---|---|
| ev-24 | holdout | 图像 | 0 | 0 | 未跑 | 同上 |
| ev-25 | holdout | 图像 | 0 | 0 | 未跑 | 同上 |

## 4. 耗时

建库是六臂共用的一次，不是每臂各建一次。`datasets/samples` 加 COCO 小样：第一次 `index_ms_total=19093`，147 块、53 个文档；第二次内容未改，`written=0`，`reused=53`，`index_ms_total=6873`。见 `reports/week4/logs/e2e_build.txt` 与 `reports/week4/logs/e2e_rebuild.txt`。

查询耗时来自 2026-10-26 的重跑，单位毫秒。A0 不编码，所以编码时间是 0。一个进程里的第一次编码含进程启动和模型加载，各次运行会差一截。dev 里 A1 的第一次编码是 1419 ms。holdout 是新进程，A4 第一次编码是 1757 ms。用来比较的是其后的中位。融合时间中位是 0.0，因为不到 1 ms。

| 臂 | 划分 | 首次编码 | 其后编码中位 | 存储查询中位 | 融合中位 |
|---|---|---|---|---|---|
| A0 | dev | 0 | 0.0 | 0.0 | 0.0 |
| A1 | dev | 1419 | 74.0 | 1.0 | 0.0 |
| A2 | dev | 386 | 40.0 | 2.0 | 0.0 |
| A3 | dev | 34 | 85.0 | 2.0 | 0.0 |
| A4 | dev | 124 | 87.0 | 3.0 | 0.0 |
| A5 | dev | 54 | 136.0 | 4.0 | 0.0 |
| A4 | holdout | 1757 | 58.0 | 9.0 | 0.0 |

这组存储查询打在 147 块的证据库上。1 万块的向量查询是另一项测量，中位 44 ms，见第 6 节。

## 5. H2

门槛是：dev 上融合臂的 Hit@5 和 MRR@10 不低于任一单通道。

A4 等于 A1（10/15，MRR 0.622222），高于 A0（0/15）和 A2（2/15）。「不低于」成立。相对最好的单通道没有正增益。

原因有两条，都没有改权重。

样本库是英文。冻结集的问句是中文。`KeywordPlan.forQuery` 只要切出中文 bigram，就只走 bigram，不会落到正文上的 trigram。所以 A0 十五题都是空名单。这不是关键词通道关掉了。核对在第 8 节，不并进上面的分数表。A3 因此等于 A1。中文文档库不是这张表的语料；`e2e_docs.txt` 里同一套关键词代码有 `keyword_terms=14915`，能和文本一起命中 ADR 0004。

图像权重是 0.2。dev 上 A2 把 ev-13 排到第 1、ev-15 排到第 3，A4 的 Top-5 仍被 `sample.pdf` / `sample.docx` 占满，这两题变成未命中。ev-11 和 ev-12 的中文问句在只开图像时也不在 Top-5，不能记成融合单独造成的失败。ev-11 的英文证据句 `A man is in a kitchen making pizzas.` 只走 clipShared 时排第 1，分数 0.003279，即 `0.2 / (60 + 1)`。这条未命中是中文问句进不了 MobileCLIP 文本塔的词表（BPE，49408 个 token，汉字 token 为 0），不是图像向量对披萨语义没有区分度。图到图用的是库里的图像向量，第 1 名是原图自己，不检验「披萨」这个词。见 `logs/e2e07_clip_diagnosis.txt`。这不改上面的 Hit@5。图像权重后来改为 0.49，记在第 10 节。

## 6. 其他数字

Chroma 对照，`reports/week4/logs/chroma_baseline.txt`：chromadb 1.5.9，`hnsw:space=cosine`，用导出的问题向量，不是产品检索器。Hit@5 0.880000 [0.700442, 0.958332] (22/25)，MRR@10 0.795714。同一脚本里 Chroma 对 numpy 的 125 个位置一致率是 1.000000。

H1 以 `reports/week4/logs/storage_equivalence.txt` 为准：125 个位置同意 125 个，`agreement=1.000000`，`passed=1`。2026-10-26 用当前两份 jsonl 重跑了 `engine/compare_ranking.py`，档案数字与输入一致（此前那份 121/125 由更早的 Chroma 输出算出，已作废）。

1 万块产品写入与查询，`reports/week4/logs/storage_capacity.txt`：SQLite 3.53.4，WAL。写入 444 ms（门槛 600），查询中位 44 ms（最小 41，最大 50，门槛 60），两项都通过。`EXPLAIN` 解释的是生产扫描语句，走 `idx_chunk_space (space=? AND dim=?)`。门槛没有改。

## 7. 已知限制

dev 只有 15 条，holdout 只有 10 条。Hit@5 从 0.67 到 0.50 的差距落在很宽的 Wilson 区间里，不能读成稳定退步。没有「库里没有答案」的题。跨页题是重复排版。图像题一共 10 条。元数据题的文件能进 Top-5，属性原文不在正文块里。A0 为 0/15，是因为问句是中文、语料是英文：冻结运行时，有中文 bigram 的查询只走 bigram，这份库的 `keyword_term` 是 0 行，而 `chunk_fts` 上的英文 trigram 能命中。A1≡A3≡A4≡A5，六臂消融在这 25 条上区分不出关键词增益。融合没有超过文本单通道，权重保持冻结。ev-11 的中文问句进不了 MobileCLIP 文本塔的词表；英文证据句只走 clipShared 时该图排第 1。E2E-07 的冻结运行仍未通过。2026-10-27 的查询改写在第 9 节，图像权重改动在第 10 节，都不替换这些数字。

## 8. 关键词路径

这是 2026-10-27 查询改写之前的核对。库是 `reports/week4/.index/evidence.sqlite`。路径来自当时的 `KeywordPlan.forQuery`，命中来自 `search_probe --arm A0`。不改冻结集，也不把这两行写进 Hit@5。日志是 `logs/keyword_path_probe.txt`。改写之后的测量在第 9 节。

库内计数：`chunk_fts_rows=96`，`keyword_term_rows=0`。同一张 FTS5 trigram 表上，`Tika` 46 行，`metadata` 6 行，`xml` 0 行。

| 问句 | KeywordPlan | A0 |
|---|---|---|
| 哪张图里有人在厨房做披萨? | bigram，11 个两字词 | `hits=0` |
| Tika | trigram | `--top 5` 返回 5 条，片段里有 Tika。FTS 全量是 46，不是这 5 条就是全部 |

中文文档库是另一份索引，不进这 25 条。`logs/e2e_docs.txt` 里 `keyword_terms=14915`。问句「产品侧用系统 SQLite 保存向量」的第 1 名是 ADR 0004，通道是 `documentText,keyword`。

# 9. 查询改写

2026-10-27 做这次改写测量时，图像权重还是 0.2。同一天稍后改为 0.49，记在第 10 节。文档文本塔仍编码原中文问句。词表命中时，MobileCLIP 文本塔改为离线英文词；没有命中则仍用原问句。中文 bigram 仍走词表，另外把长度不少于 3 的拉丁串和这些英文词送进 trigram。整句中文不送进 trigram。同一块保留较高分。词表是仓库里的固定视觉词，不是证据句，也不是 jieba。日志是 `logs/query_rewrite_probe.txt`。

只开图像（A2）时，五条图像问句的金标文件都进了 Top-5。当时融合（A4）的 Top-5 里没有这些图像文件。图像块正文是空的，关键词打不中图像行。

| 问句 | 送给 CLIP 的字符串 | A2 金标名次 | 冻结 A2 |
|---|---|---|---|
| 哪张图里有人在厨房做披萨? | person kitchen pizza | 1，`000000397133.jpg` | 未进 Top-5 |
| 哪张图里有一辆蓝色自行车停在路边? | blue bicycle parked roadside | 1，`000000174482.jpg` | 未进 Top-5 |
| 哪张图里的浴室淋浴墙是破的? | bathroom shower wall broken | 2，`000000403385.jpg` | 1 |
| 哪张图里有大型客机停在机场跑道上? | large airliner parked airport runway | 1，`000000348881.jpg` | 9 |
| 这张图片版面顶部的文档标题写的是什么? | picture layout top document title | 1，`sample.png` | 3 |

浴室那条从冻结的第 1 名变成第 2 名，仍在 Top-5 里。`sample.png` 排到第 1 只说明文件被召回，不说明模型读出了版面上的标题。

关键词：四条场景问句 `keyword_hits=0`。含「文档」「标题」的问句命中 `sample.pdf` / `sample.docx`，没有命中 `sample.png`。离线 的探针仍是空的，命中数 0。说明里的 `Tika` 没有词表命中，CLIP 字符串保持原问句，trigram 探针只有 `Tika`，前 5 条都是 `sample.docx`。权重仍是 0.2 时，A4 的五条 Top-5 都是这两份说明；标题问句的第 3 名和第 5 名片段是 `Apache Tika Documentation`，通道是 `documentText,keyword`。

# 10. 图像权重

文本权重仍是 0.5，关键词仍是 0.3，`k` 仍是 60。图像从 0.2 改为 0.49。`parameters_changed_after_freeze=1`。0.49 取自名次公式：大于 `0.5 * 62 / 64` 时，图像单独第 2 名进得了 Top-5；小于 0.5 时，文本单独第 3 名不会被同分路径序挤出。没有用 dev 标签搜索这个数。第 2 节和第 5 节的日志不改。新日志是 `logs/retrieval_bench_reweight_dev.txt`、`logs/retrieval_bench_reweight_holdout.txt`。holdout 这是第二次查看，不替换原来的 `retrieval_bench_holdout.txt`。0.900 的口径写在本节末段。

dev 上 A1 仍是 10/15，MRR 0.622222。A2 是 5/15，MRR 0.300000，五条都是图像题。A4 是 14/15，MRR 0.720741，高于 A1。A0 是 5/15，题号是 ev-04、ev-07、ev-08、ev-09、ev-10，图像题仍然是 0。ev-07 和 ev-09 的探针是词表里的 `document`，第 1 名是 `sample.docx` 的 `0–25`，正文是 `Apache Tika Documentation`，那是文档标题行，不是「说明正文」；同一问句的第 2 名 `178–262` 才是正文里的 `This documentation serves as an example and a starting point`。按文件算 Hit@5 记 1 成立，但 ev-07 与 ev-09 的首位命中来自标题行，写报告时不能把它当成正文命中。ev-10 另有探针 `text`，打在正文的 `metadata and text` 上；探针 `title` 在库里 0 命中。ev-04 的探针是问句里的 `Pipes`，第 1 名是 `Pipes — process-isolated` 那一行。`Pipes` 共 5 块：三块是正文里的 `Pipes — process-isolated, fault-tolerant extraction at scale`（docx 一块、pdf 两页各一块），两块是 pdf 页脚链到 `pipes/index.html` 的 URL。首位是正文，后面有页脚。ev-08 的探针是问句里的 `commit`，命中 `Built from commit: 47c1a0335`，3 块。`Pipes` 和 `commit` 不是词表。ev-01 到 ev-10 的 A4 仍全部命中，ev-09 从冻结时的第 3 名变成第 2 名。

冻结集的文本和元数据题是 15 条：dev 10 条、holdout 5 条。12 是 dev 这 10 条、50 个 Top-5 位置里的图像行，落在其中 6 条上，每条 2 行。冻结 A4 的这 50 个位置没有图像行。权重 0.49 之后是 12 行：ev-01、ev-02、ev-03、ev-05、ev-06 的第 3 名和第 5 名是图像，ev-08 的第 4 名和第 5 名是图像。ev-04、ev-07、ev-09、ev-10 的 Top-5 仍全是说明文件。金标文件仍在 Top-5 里，Hit@5 没有下降。图像单独第 1 名是 `0.49/61`，介于文本单独第 2 名 `0.5/62` 和第 3 名 `0.5/63` 之间，所以只走文本的块会把第 3 名让给图像。单测把前五名锁成文本第 1、文本第 2、图像第 1、文本第 3、图像第 2。ev-02 没有词表命中，Top-5 里仍有图像行。holdout 那 5 条里，ev-16、ev-18、ev-19 另有 6 行，不并进这 12，也不拿去调权重。

| 题 | A4 名次 | Top-5 里的金标 |
|---|---|---|
| ev-11 披萨 | 3 | `000000397133.jpg` |
| ev-12 蓝自行车 | 3 | `000000174482.jpg` |
| ev-13 破损淋浴墙 | 5 | 金标 `000000403385.jpg` 在第 5 名。第 3 名是另一张图 `000000006818.jpg` |
| ev-14 客机 | 3 | `000000348881.jpg` |
| ev-15 版面标题 | 9 | 未进 Top-5。Top-5 是 `sample.docx` / `sample.pdf`，通道含 keyword |

ev-13 的 Top-5 里有两张图。图像行名额留 2 张时，第 3 名和第 5 名都在，这一题不变。名额留 1 张时留下第 3 名 `000000006818.jpg`，金标 `000000403385.jpg` 出 Top-5。

词表有 128 个中文键、117 个不同的英文词。其中 30 个英文字符串与 COCO 80 类的类别名相同。`orange` 在词表里是颜色，与 COCO 的水果类同名。`sofa`、`phone`、`ball`、`table` 和 COCO 的 `couch`、`cell phone`、`sports ball`、`dining table` 不是同一字符串。10 条图像题都命中词表，holdout 也在内。dev 上是「人 / 厨房 / 披萨」「蓝色 / 自行车 / 停在 / 路边」「浴室 / 淋浴 / 墙 / 破的」「大型 / 客机 / 停在 / 机场 / 跑道」「图片 / 版面 / 顶部 / 文档 / 标题」。holdout 上是「白色 / 房间 / 桶」「一群 / 山羊 / 人」「车 / 停在 / 楼」「长椅 / 水面 / 船」「图片 / 版面」。`山羊`、`桶`、`客机`、`路边`、`破的` 不是 COCO 类别名，它们出现在冻结问句里。15 条文本和元数据题里有 5 条命中：ev-01 的「人」，ev-07、ev-09、ev-10、ev-20 的「文档」，ev-10 还有「标题」「文字」。词表没有从 25 条英文 `evidenceText` 抄句子。冻结问句的措辞和词表键有重合，holdout 问句也在其中。改写之后的 holdout 命中，测的是这些已经覆盖的说法。这段只记录现有词表。

holdout 的 A4 是 9/10，MRR 0.620952。ev-16 到 ev-20 仍是第 1 名。ev-21 第 3，ev-22 第 3，ev-23 第 5，ev-24 第 5。ev-25 是第 7，Top-5 里没有 `sample.png`。版面上的字仍靠文本文件的正文和关键词，图像块正文是空的，0.49 盖不过两路都命中的说明块。产品检索的 holdout A4 已经查看 2 次：冻结日志是 5/10 = 0.500，这一次是改完查询和权重之后的重测。0.900 不能与 0.500 并列当成提升，也不再是无偏估计。冻结系统的那一次仍是 `retrieval_bench_holdout.txt`。本日志的 `holdout_scored_runs=1` 只计这一次进程。Chroma 对照排过这些查询向量，那是另一条存储路线，不计进这两次。